



Tire wear particles enhance horizontal gene transfer of antibiotic resistance genes in aquatic ecosystems

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ABSTRACT

Microplastics (MPs) have introduced new surfaces for biofilm development and gene exchange among bacteria. We investigated Tire Wear Particles (TWPs) for their involvement in horizontal gene transfer (HGT), particularly in relation to associated metals in the matrices of TWPs. We employed red-fluorescently tagged *E. coli* strain as a donor with green-fluorescently tagged, broad-host-range plasmid pKJK5, resistant to trimethoprim. As a recipient, we utilized *Pseudomonas* sp. and a natural lake microbial community. HGT activity on TWPs was determined and compared with that on polystyrene (PS) (with and without metals), and chitosan, which was used as a natural surface. Exposure to TWPs significantly enhanced HGT frequency of antibiotic resistance gene (ARG) from donor to recipient compared to PS and chitosan, and metals of TWPs further promoted HGT. HGT frequency on TWPs with *Pseudomonas* sp. was found to be 10^{-3} at 30 °C. in the lake community, it was similarly high already at 25 °C suggesting a higher permissiveness of the natural microbial community towards ARG at lower temperatures. This study sheds light on the potential impact of TWPs in promoting HGT, forming the basis for health risk assessments of TWPs and more generally of MP pollution in various aquatic ecosystems.

1. Introduction

Despite being a global problem, plastic pollution is increasing due to a steady increase in global consumption from 1.7 million tons (MT) in 1950 to 380 MT in 2020 (PlasticsEurope, 2021). A significant portion of this large volume of plastic (around 0.5%) subsequently makes its way into the environment, where it accumulates, especially in aquatic ecosystems (Ritchie and Roser, 2023; Abdolapur Monikh et al., 2023). This has far reached consequences when it is fragmented into smaller pieces, known as microplastics (MPs: 1 µm - 5 mm size) and nanoplastics (size <1 µm) (Thompson et al., nd; Maharana et al., 2020; Sarkar et al., 2022, Jaffer et al., 2023, Mandar Bandekar et al., 2022). Fragmentation in nature is considered the largest source of MPs. This includes, among others, abrasion of tires which can lead to the release of MPs (Yadav et al., 2023) known as Tire Wear Particles (TWPs). TWPs are estimated to be one of the largest contributors to plastic pollution in the

environment with their relative contribution estimated to be 5–10% of all MPs in oceans (Birch et al., 2020; Jan Kole et al., 2017). The global annual emissions of TWPs are estimated to be approximately 3.4 million tons, with Europe contributing annually around 1.3 million tons and the United States between 1.1 and 1.8 million tons (Baensch-Baltrusch et al., 2020; Jan Kole et al., 2017). Unlike other plastic types, TWPs are composed of a variety of materials including synthetic rubber, filling agents, vulcanization agents, and other additives (Wik and Dave, 2009; Aatmeeyata & Sharma, 2010). As a result, TWPs can be toxic both to the environment and organisms (Cunningham et al., 2022).

Previous studies demonstrated that once MPs are present in aquatic ecosystems, a conditioning biofilm appears on their surfaces within a few hours to days (Nguyen et al., 2023; Rozman et al., 2023), typically within 8–12 days (Tu et al., 2020). This subsequently attracts highly diverse and metabolically complex microbial colonizers (e.g., toxic, pathogenic, or invasive species) with the resulting biofilms consisting of

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microorganisms such as bacteria, fungi and algae (Bakker et al., 2003; Salta et al., 2013; Kettner et al., 2019; Yang et al., 2020). The bacterial communities formed on these MPs are different from the surrounding water and other natural surfaces like wood (Zettler et al., 2013; Oberbeckmann et al., 2016; Kettner et al., 2017; Arias-Andres et al., 2018a). In this context, MPs represent surfaces for niche partitioning and microbial competition (Sørensen et al., 2005a; Ellis et al., 2015; Arias-Andres et al., 2018a). Since MPs can be transferred over long distances in aquatic ecosystems, horizontally and within water columns (vertically), post biofouling event, they may act as efficient vectors for spreading pathogenic and other harmful microbes (Kirstein et al., 2016; Rodrigues et al., 2019). For example, wastewater treatment plants are significant sources of MP emissions into freshwater systems and provide environments where infectious and antibiotics resistant bacteria (Numberger et al., 2019) presumably colonize MP surfaces. Once released, these MPs can disperse extensively across various ecosystems, carrying the attached bacteria on their surfaces.

However, transfer of bacteria from one ecosystem to another is not the only hazard associated with microbial MP colonization. MP surfaces were demonstrated to facilitate HGT due to increased nutrient availability and high microbial cell densities (Arias-Andres et al., 2018b). Several studies have addressed the effect of MPs on accumulation and persistence of microbes that carry antibiotic resistance genes (ARGs), particularly in contexts such as maricultural (Arias-Andres et al., 2019; Zhang et al., 2020) and activated sludge systems (Pham et al., 2021). This phenomenon has gained significant momentum, particularly after our previous study revealed that MPs can serve as a platform for the transfer of ARGs from pathogenic bacteria to the broader environmental bacterial community. We observed an increased frequency of such HGT between bacteria within biofilms formed on MPs compared to planktonic water (free-living) and natural aggregate associated microbes (Arias-Andres et al., 2018b). This raises a serious concern that MPs might potentiate antibacterial resistance in colonizing pathogenic bacteria and that these ARGs could increasingly spread via HGT to the environmental bacterial community.

Unlike other MPs, TWPs are additionally loaded with high concentrations of heavy metals such as Zn, Al and Pb (Sadiq et al., 1989; Kreider et al., 2010). These metals can play a significant role in facilitating HGT in several ways such as inducing stress responses which may increase the frequency of HGT by promoting a state in bacteria that makes their genetic material more accessible or susceptible to exchange, contributing to the co-selection of antibiotic resistance genes, and inducing gene mutation (Seiler and Berendonk, 2012; Baker-Austin et al., 2006). There is no study investigating how TWPs and the metals present in their matrix can influence HGT in biofilm on the surface of the particles. We hypothesize that the presence of metals in TWPs can further elevate HGT of ARGs from donor to environmental recipient bacteria compared to other MP types and to natural surfaces such as chitosan. Nevertheless, one must consider that metals are present in the structure of the particles and not solely in a free form. Thus, physicochemical properties of the particles including surface area and chemical composition as well as interactions between bacteria and MP might play a significant role in metal release dynamics from the particles and, in turn, their effect on bacterial HGT.

Therefore, the main objective of this study is to understand the influence of TWPs on bacterial HGT of ARGs. To achieve this, we utilized two model plasmid transfer systems. A red-fluorescently tagged *E. coli* strain with the self-transmissible, green-fluorescently tagged, plasmid pKJK5, encoding resistance to trimethoprim, in combination with either a single strain *Pseudomonas* sp. or a complex natural lake microbial community as recipients to determine HGT (plasmid) frequency under five different conditions. The concept is that the green fluorescence protein is repressed in donor cells while active upon plasmid transfer in the transconjugant cells (bacteria incorporating the plasmid via conjugation). Accordingly, the number of donors (red) and transconjugants (green) can be determined based on fluorescence using flow cytometry

(FCM) and fluorescent microscopy and thus quantification of HGT. We applied chitosan as a model of a natural surface to compare potential TWPs effects to those caused by naturally occurring particulate matter for microbial biofilm formation (Corno et al., 2015; Köllner et al., 2012). In the absence of standard material, we used polystyrene (PS) MPs of 50 μm size as a representative for TWPs in their "pure" state, devoid of additional components such as metals.

2. Materials and methods

2.1. Materials

Milli-Q (MQ) water was produced by reverse osmosis and purified by a Millipore MQ system. For this study we used one of the most commonly used automobile tires in Germany as a tire model. PS MPs were purchased from ThermoFisher Scientific (Waltham, USA) in a size range of 50 μm . AlCl_3 , MnCl_2 , CoCl_2 , CuCl_2 , ZnCl_2 and Cd were obtained from Sigma-Aldrich (Germany). Chemicals used for DEV medium were meat peptone (Art.-Nr 2366.3: Carl Roth GmbH + Co. KG, Germany), meat extract (Sigma-Aldrich, Germany, Lot: BCBR4306V) and NaCl (Th. Geyer, CAS No. 7647-14-5). Ethanol 90% (Technisolv, VWR chemicals, Belgium: CAS No. 64-17-5) was used as flow cytometer carrier fluid. MF-Millipore™ Membrane filters with 0.22 μm pore size and 47 mm diameter (Millipore, USA) were used to concentrate bacteria for microscopy.

2.2. TWPs preparation

TWPs were obtained by scrapping and shaving the tire treads using metallic scrubbers following the top-down approach reported previously after some modification (Abdolahpur Monikh et al., 2020). Subsequently, the scraped particles were subjected to a washing process using autoclaved MQ water to avoid organic matter contamination from the used tires. 10 ml of water was used to wash 1 g of TWPs and the washing was repeated 3 times. TWPs were then sieved through stainless steel sieves with pore sizes of 100 μm and subsequently 10 μm , resulting in the isolation of TWPs ranging in size from 10 to 100 μm . The particle size of all TWPs was controlled within the environmentally relevant range of 10–100 μm (Kovochich et al., 2021; Lopez et al., 2023). This not only aligns with environmental conditions but also facilitated the investigation of biofilm formation. To remove soluble or lightly bound organic matter and dust contaminants, the particles underwent additional washing with MQ water. Sterilization was achieved by washing the particles with 70% ethanol and subsequently drying them overnight at 50 °C. The sterilized TWPs were then stored in a glass bottle at room temperature for further use.

2.3. Characterization of TWPs

2.3.1. Determining TWPs polymer type

To identify the polymers in TWPs, the samples were analyzed using Fourier transform infrared (FT-IR) spectrometry following previous method (Uurasjärvi et al., 2020). A Thermo Nicolet iS50 FT-IR spectrometer was used in ATR mode, with spectral resolution of 4 cm^{-1} and 32 scans.

2.3.2. Quantification of metals in TWPs

The TWPs (3.54 ± 0.75 mg/10 mL, mean $\pm$ SD) were digested in HNO_3 (70%) using a water bath at 65 °C for 2 h (Abdolahpur Monikh et al., 2020). The digest was centrifuged at $10,000 \times g$ for 20 min (Sigma Lab. Centrifuges GmbH, SN 7101911999). The supernatant was mixed with 5% HNO_3 for further analysis. Metal concentrations in the suspensions were measured using inductively coupled plasma mass spectrometry (ICP-MS) (PerkinElmer NexION 350D) operating in standard mode. The conditional setup of the ICP-MS is given in supporting information (Table S1).

2.4. Metal release and toxicity assessment from TWP

The leaching of metals from TWPs was measured over 48 h at 30 °C. The composition and concentration of leached metals were analyzed at intervals of 0, 12, 24, 36 and 48 h. The concentration of metals used for metal release was 60 mg/L. Details are provided in supporting information (Text S1). Subsequently, the effect of leached metals on the growth of bacteria was determined. A toxicity test was conducted using a mixture of metals found in the TWP leaching experiment (Al, Mg, Co, Cu, Zn, and Cd). Bacterial growth inhibition was measured by changes in bacterial densities, determined via absorbance at 600 nm after 24 h of incubation. Overnight bacterial cultures were inoculated into fresh media containing metal mixtures at concentrations of 0.01, 0.1, 1, and 10 mg/L. Each metal concentration was tested in triplicates, along with a control, and bacterial growth was monitored using a UV-Vis spectrophotometer. The donor and recipient bacteria were exposed separately to the leached concentration over 24 h. The detailed information is provided in supporting information (Text S2).

2.5. Effects of TWPs on horizontal ARG transfer

2.5.1. Strains and cultures

E. coli MG1655 tagged chromosomally with a *lacIq-pLpp-mCherry-Km^R* gene cassette, which confers red fluorescence, kanamycin resistance and a *lacIq* repressor, was used as the plasmid donor (Klümper et al., 2014a, 2014b, 2017). The donor hosts the IncP-1 ϵ broad host range (BHR), conjugative plasmid pKJK5:*gfpmut3b* that confers trimethoprim resistance (Bahl et al., 2007). The donor represses the production of plasmid-encoded *gfpmut3b* through the *LacI* repressor, but *gfpmut3b* is expressed upon conjugative transfer allowing for the quantification of green fluorescent transconjugant cells (Sørensen et al., 2005b). Based on this principle, the plasmid transfer frequency in the presence and absence of TWPs was determined. As recipient strain *Pseudomonas* sp., previously isolated from Lake Stechlin (Germany) and shown to be able to take up plasmid pKJK5, was used (Arias-Andres et al., 2018b). In addition, natural microbial communities of Lake Stechlin, Germany were also used as recipients. All strains and communities were cultured in nutrient broth (DEV) and nutrient agar (DEV). Antibiotics (Kanamycin Km 50 μ g/mL, Trimethoprim TMP 30 μ g/mL) were supplemented to support the donor strain pure cultures. The donor and recipient did not show any negative effect on each other's growth (Fig. S1).

2.5.2. Optimization of the conditions for HGT tests

To evaluate the optimal conditions under which HGT occurs readily, three factors, temperature, incubation time and donor to recipient (D/R) ratio, were evaluated. All experiments were conducted using pre-sterilized 100 mL flasks as microcosms, containing 20 mL of DEV nutrient broth in each flask. The donor and recipient bacteria were added in respective ratios (see below) to achieve a total bacterial density of OD₆₀₀ = 0.05 in the mating medium at the start of the experiments. Experiments were conducted in dark conditions under constant agitation at 150 rpm. The effect of temperature on HGT was tested by incubating TWPs (60 mg/L) with donor and recipient at a fixed ratio of 1:10 at 17, 20, 25, 27 and 30 °C for 24 h. To examine the effect of incubation time, TWPs (60 mg/L) were incubated over 24 h and 48 h with donor and recipient at a ratio of 1:10 at 30 °C. To test the effect of the D/R ratio on HGT frequency, five different ratios of donor and recipient cells (1:100, 1:10, 1:1, 10:1, 100:1) were selected. Five replicates were prepared for each treatment. A control treatment without TWPs was also set up for each temperature.

Aliquots of the liquid (planktonic phase) and biofilm samples (particle phase) formed on the particles were taken from each replicate at the end of the experiment. Biofilm formed on the surface of particles was detached by first carefully pipetting the particles into 5 mL Eppendorf tubes and washing them twice with a 0.9% sterile NaCl solution to

remove any planktonic bacteria. Thereafter, the particles were allowed to settle and the NaCl solution was replaced with 2 mL of a sterile mixture of sodium pyrophosphate (50 mM Na₄O₇P₂) and Tween 80 (0.05%). The tubes were then incubated for 30 min and vortexed three times for 30 s intermittently. The detachment of the biofilm was completed with a sonication step at an intensity of 60 W for 30 s, which was repeated three times with intermittent hand-shaking for 30 s. Both types of samples, particle and planktonic phase were then subject to flow cytometric analysis as outlined below.

2.5.3. Performing HGT on TWPs under optimum conditions

After determining the optimum HGT conditions, the experiments were carried out as described above under five different conditions. These included a) TWPs at 60 mg/L concentration; b) PS at 60 mg/L concentration; c) metals without TWPs equivalent to the previously determined amount leached from TWPs, all in the range of $\times 10^{-6}$ mg/L (Al = 0.74, Mn = 6.24, Co = 0.003, Cu = 0.069, Zn = 0.31 and Cd = 0.0011), d) Chitosan at 60 mg/L concentration, and e) a control containing exclusively nutrient broth DEV. The PS represented the TWPs without metals to differentiate between any effect of TWPs vs. metals. PS provided a smooth and uniform surface as compared to polyethylene (PE) or polypropylene (PP) that was necessary to avoid the influence of structural irregularities on biofilm formation (Pan et al., 2022; López et al., 2024). The pure form of PS was used to avoid the confounding influence of other additives found in TWPs. In addition, there is no metal-free standard available for TWPs, we applied PS MPs as a standard because TWPs consist of styrene as one of the polymer components as determined by FT-IR (Fig. S2).

We also conducted a follow-up experiment to determine whether the metals associated with particles or the readily available leached metals in the suspension phase played a role in HGT. In this setup, we introduced an additional treatment by adding PS (60 mg/L) and metals (PS + Metals) to the existing treatments (TWPs, PS, metals, and chitosan). A control containing only nutrient broth (DEV) was also included. The metal composition and concentration used were the same as in the previous experiment.

The concentration of TWPs (60 mg/L) was chosen as the maximum environmental concentration predicted by a previous study (Wik and Dave, 2009) after performing the meta-analysis on tire particles in all environmental compartments, including air, water, soils/sediments, and biota. The treatment with chitosan was used to mimic natural surfaces in aquatic environments (Corno et al., 2015; Köllner et al., 2012), which is well colonized and utilized by bacteria, and compare the effects of naturally occurring and anthropogenically introduced particles. Sample from the suspension and particle associated biofilm were taken at the end of the experiment as described above and analyzed through flow cytometry.

2.6. Evaluation of HGT frequencies through flow cytometry

The ratio of donor, recipient and transconjugant cells from the liquid broth and the particle phase from each sample of the above-described experiments were analyzed by flow cytometry (FACSAriaII instrument) and BD FACSDiva TM software v6 (Becton Dickinson Biosciences, San Jose, CA). The HGT frequency was expressed as T:R (transconjugants/total recipients) as the recipient cells formed a major pool of cell numbers after optimizing the ratio of donor and recipient (D:R) (Ma and Bryers, 2013). The flow cytometer was equipped with a 488 nm laser (100 mW) that was connected to a green fluorescent detector with a range of 500–550 nm and a 532 nm laser (150 mW) that was connected to a red fluorescent detector with a range of 600–620 nm. The side scatter threshold was set at 300 (Gasol and Del Giorgio, 2000; Arias-Andres et al., 2018b). Samples were diluted with phosphate buffer saline (PBS) to attain an event rate in the range of 200–500 cells per second. Total bacterial numbers were analyzed by setting a gate for both strains on a bivariate FSC-A versus SSC-A plot. Gates for each donor and

recipient were set on a bivariate FITC-A (green fluorescence) versus PE-Texas Red-A (red fluorescence) plot, using cell suspensions from each strain whereas for gating the transconjugants IPTG induced cells were used (Sørensen et al., 2003). This allowed quantification of total bacterial numbers, donors, recipients and transconjugants for each sample. The detection limit was set at 15,000 cells. The flow cytometer was also used to analyze blanks, which consisted of PBS and sterile filtered (0.2 µm) and autoclaved samples. These blanks were processed and analyzed in the same way as the samples, allowing for noise correction of cell counts.

2.7. HGT in a lake microbial community

To evaluate the impact of TWPs on HGT frequency in a complex microbial community, water samples were taken from the surface of Lake Stechlin, Germany. The microbial community in water was gradually acclimatized to the desired temperatures (20 and 25 °C). These temperatures represent the lower and higher temperatures of the lake. TWPs (60 mg/L) were incubated with *E. coli* (donor) and the lake microbial community (recipient) in the ratio of 1:10 at 20 °C and 25 °C for 24 h. DEV nutrient broth was added as an additional nutrient source in a ratio of 1:1000 to the water sample. The addition of DEV nutrient broth to water samples provided sufficient nutrients to support microbial growth during acclimatization and the incubation period. This approach has been used in previous studies where low concentrations of nutrients were added to ensure microbial viability without drastically changing the microbial ecology or experimental conditions (Bussmann et al., 2001; Azmuda et al., 2019). Treatment without TWPs acted as a control. Samples were taken as described above and analyzed using flow cytometrically.

The biofilm developed during the experiments was visualized using both a fluorescent microscope and scanning electron microscope (SEM). Detailed methodologies are provided in the supporting information (Text S3).

2.8. Data and statistical analysis

All statistical analyses were performed in R version 4.2.1 (2022-06-23 ucrt). The transconjugant-to-recipient (T:R) ratios were calculated for both particle and planktonic phases in all experiments for each treatment. To compare T:R in particle and planktonic phases between different treatments, a Kruskal-Wallis non-parametrical test was used followed by Dunn's pairwise comparison. The Mann-Whitney-Wilcoxon Test was used to compare T:R ratios within a treatment between water and particle phases, or to compare the transfer frequency in one

treatment with other treatments. In this study, linear regression analysis was used as the primary statistical tool to investigate the interactions between key variables influencing horizontal gene transfer (HGT) on tire wear particles (TWPs).

3. Results

3.1. Composition and metal release from TWPs

The studied TWPs mostly consisted of styrene-butadiene rubber (SBR) based on the absorption spectra obtained through FTIR (Fig. S2). TWPs further contained a range of metals with Zn exhibiting the highest concentration (73441.89 ± 12472 mg/L), followed by Si, Al, Fe, Mg, Pb, Cr, Cu, Mn, Co, and Cd in decreasing order of concentration (Fig. 1a) (Table S2). The leaching of metals from TWPs into the mating medium was examined for 48 h and revealed that Al, Mn, Co, Cu, Zn, and Cd were among the metals that leached into the nutrient broth. The concentration of these leached metals displayed a distinct pattern over time (Fig. 1b). Initially, there was an increase in the total concentration after 12 h, which was followed by a relatively stable phase for up to 36 h. Subsequently, a gradual decline in concentration was noted (Fig. 1b) (Table S3).

The metals released over exposure time (24 h) had no effect on the viability or growth dynamics of bacteria (Text S4, Fig. S3).

3.2. Colonization of TWPs

The colonization of TWPs by *E. coli* and *Pseudomonas* sp. was observed using SEM. SEM images (Fig. 2a and a') confirmed that a biofilm developed on the surface of TWPs during 24 h of incubation at 30 °C. We further assessed bacterial colonization on PS (used as a control) (Fig. 2b and b') and chitosan (Fig. 2c and c'), where the formation of biofilms on the surface of both materials was confirmed.

3.3. Optimizing the conditions for HGT assays

3.3.1. Donor-to-recipient (D/R) ratios

First, we investigated the effect of D/R ratios on the HGT frequencies to optimize and determine the appropriate ratios of both donor and recipient cells during conjugation assays. We measured the transfer frequency of HGT on TWPs in different ratio treatments (1:100, 1:10, 1:1, 10:1, 100:1) by calculating the T:R (recipients include the total number of *Pseudomonas* sp.). A significant difference in HGT frequency among different ratios was determined based on the Kruskal-Wallis test (p -value = 0.0006) (Fig. 3a). The T:R ratio showed a significantly higher

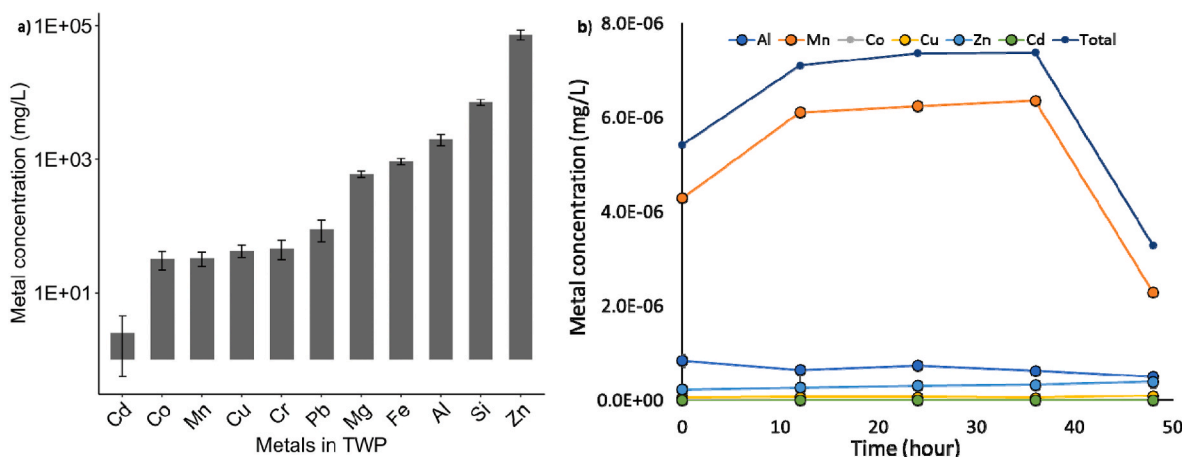


Fig. 1. a) Concentration of heavy metals (mg/L) present in tire wear particles (TWPs). b) Metals released over time from TWPs (mg/L, N = 3) into the nutrient broth used for bacterial culture over 0, 12, 24, 36 and 48 h detected by ICP-MS. Co and Mn having similar concentrations and thus Co isn't visible in the graph.

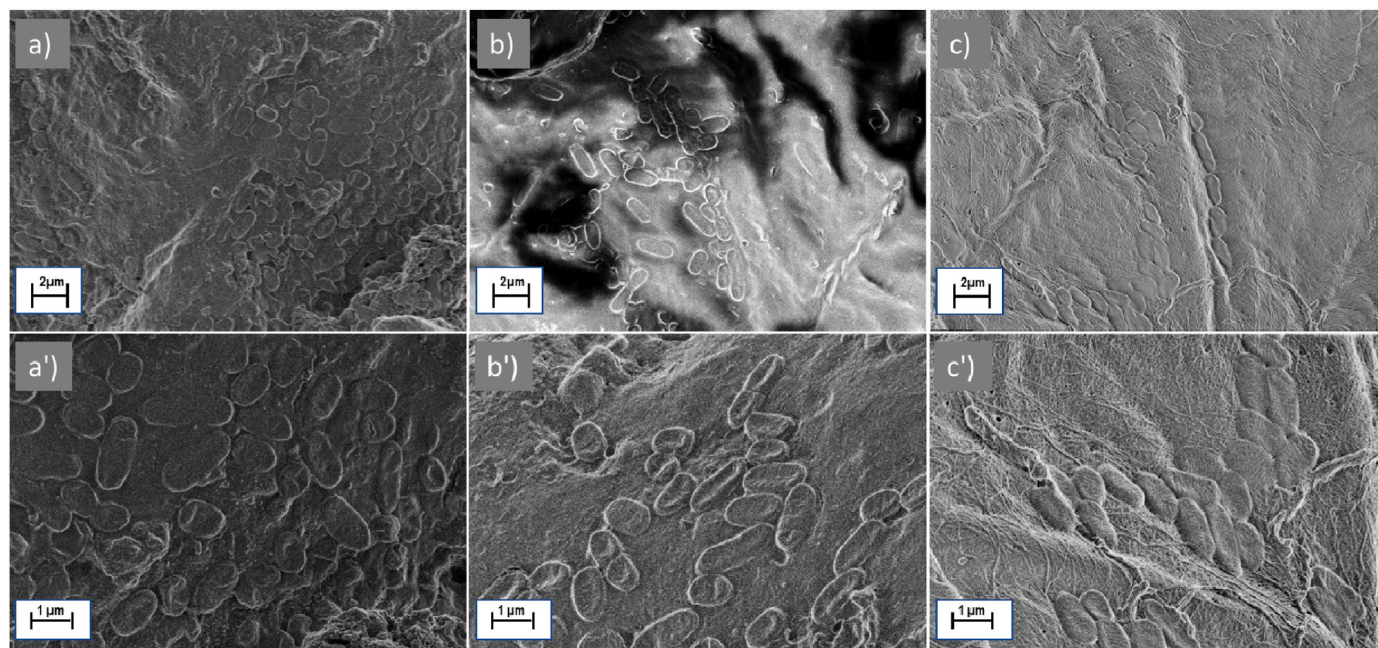


Fig. 2. SEM images confirming biofilm formation on (a) TWPs (b) PS and (c) chitosan after 24-h incubation with donor and recipient bacteria. Images in the top row (a, b, c) were captured at 5 KX magnification, while those in the bottom row (a', b', c') were captured at 10 KX magnification.

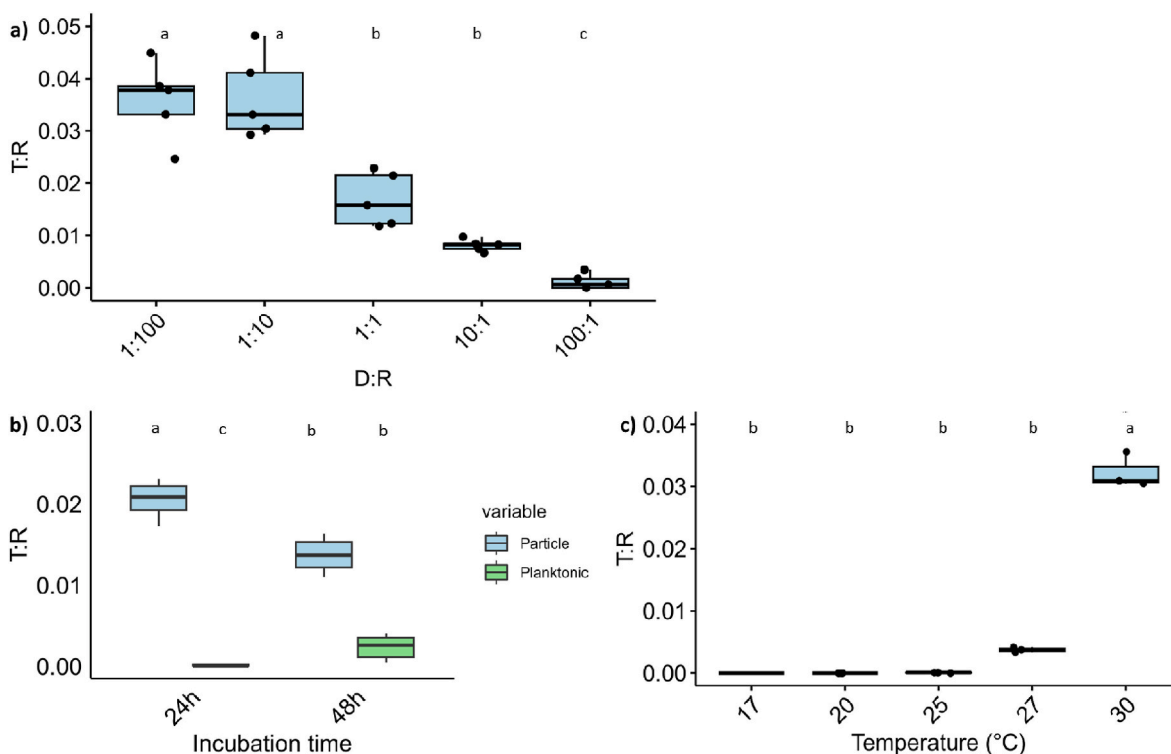


Fig. 3. Box plots represent T:R at different (a) D/R ratio, (b) incubation time and (c) temperature. Error bars represent the standard deviation from the measurements. 'a', 'b' and 'c' indicates significantly different groups in the transfer frequency between treatments (Post hoc Wilcoxon test, $p < 0.05$).

HGT frequency in the 1:10 and 1:100 proportions compared to all other tested ratios. However, no significant difference was found between the donor and recipient ratios of 1:10 and 1:100 based on the Wilcoxon pairwise comparison test (p -value = 0.8413). The HGT frequency (T:R) in the 1:10 and 1:100 ratios were approximately 27 times higher than the lowest frequency observed in the 100:1 treatment.

Upon comparing the proportion of donor (*E. coli*) cells on particles,

no significant difference was observed among different ratios. However, a significantly higher number of recipients (*Pseudomonas* sp.) was observed on particles in the 1:10 (D/R) treatment than all other tested ratios. When donor and recipient strains were incubated with TWPs separately, the number of cells/mg of TWPs were 15850 ± 2209 (mean $\pm$ SD) and 163578 ± 54007 (mean $\pm$ SD), respectively.

3.3.2. Incubation time and temperature

There was a significant difference in HGT frequency (T:R) between 24 and 48 h of incubation in the particle phase with the highest HGT frequency recorded after 24 h (Wilcoxon pairwise comparison test, $p = 0.007$) (Fig. 3b). In contrast, the HGT frequency (T:R) in the planktonic phase revealed a higher number of transfer events after 48 h of incubation (Wilcoxon pairwise comparison test, $p = 0.007$) (Fig. 3b). The D/R in planktonic phase and on the surface of particles after 24 h of incubation was 0.1 and 0.07, respectively. However, this ratio changed to 1.2 and 0.003, respectively, after 48 h of incubation. Notably, after 48 h, there was a significant decrease in the number of donor cells (87.26%), while the number of recipient cells increased by 2.8 times on the particles.

In general, we found that higher temperatures resulted in a significant increase in the number of conjugation events (T:R) with the highest HGT frequency at 30 °C when compared to all other temperatures tested (17, 20, 25, 27, and 30 °C) (Kruskal-Wallis test, $H = 13$, $p = 0.01$) (Fig. 3c). The HGT efficiency was found to be significantly higher at 30 °C, compared to both 27 °C ($p = 0.04$) and 25 °C ($p = 0.04$), which was approximately 8.6 and 547 times greater, respectively. Although bacterial growth was noticed at 17 °C and 20 °C, plasmid exchange was basically absent. Furthermore, the process of conjugation was not observed in the planktonic stage, except when conducted at 30 °C. Yet, even at 30 °C, the conjugation in the planktonic phase revealed a 2.6-fold decrease compared to those on the particle. Interestingly, comparing the overall plasmid exchange frequency at 30 °C to the control showed a 17-fold higher. This suggests that despite a decrease of HGT frequency in the planktonic phase, HGT was significantly increased at 30 °C compared to control conditions. As the HGT frequency in the control was relatively lower or absent, it has not been shown in the figures.

Thus, for HGT a donor: recipient ratio of 1:10 or 1:100, 24 h incubation time and 30 °C were found to be optimum.

To explore potential interactions between temperature, incubation time, and the donor-to-recipient ratio on HGT, we applied a linear regression model. Based on our analysis, none of the interaction terms reached statistical significance (p -value > 0.05) (Table S4). The interaction coefficients were estimated as 0.0001 for Incubation Time * Temperature, 0.0144 for Incubation Time * Ratio, -0.00218 for Temperature * Ratio, and -0.0012 for Incubation Time * Temperature * Ratio. While these results suggest that the combined effects of these variables may not have a strong influence on HGT in this specific model, further investigation or alternative analytical approaches may be necessary to confirm these findings (Table S4).

3.4. TWPs act as a hotspot for horizontal gene transfer

The optimal conditions for HGT were identified as a 1:10 or 1:100 D/

R ratio, 24-h incubation time and 30 °C temperature. Using fluorescence microscopy, we confirmed that gene transfer did occur on the surface of the TWPs (Fig. 4). There was a significant difference in HGT frequency between conditions with and without TWPs ($p = 0.007$). In the presence of TWPs, the overall average HGT frequency was $2.347 \pm 2.45 \times 10^{-3}$ (mean $\pm$ SD). However, without TWPs, we didn't observe any gene transfer events. HGT was further investigated in two different environments: on TWP's surface ($4.5 \pm 1.3 \times 10^{-3}$, mean $\pm$ SD) and in the surrounding suspension (planktonic phase) ($0.168 \pm 0.25 \times 10^{-3}$, mean $\pm$ SD) (Fig. 5). We found a significant difference ($p = 0.011$) between these two phases. This was also confirmed by fluorescence microscopy (Fig. 4). Plasmid transfer activity on particles ($4.5 \pm 1.3 \times 10^{-3}$, mean $\pm$ SD) was approximately 27 times higher than in the planktonic phase.

In order to see the influence of the chemical composition of the particles on HGT, we compared the results obtained for TWPs with PS and with chitosan (as natural organic surfaces). There was a significant difference in HGT frequency among all treatments with TWPs showing the highest HGT (Fig. 5). HGT frequency was below the limit of detection on the particle and in the planktonic phase of the PS treatment (Table S5). There was a statistically significant difference in HGT frequency between the planktonic phases of treatments containing TWPs, PS, metals, chitosan, and the control (Kruskal-Wallis test, $H = 22.58$, $p = 0.00015$) with highest HGT found in the TWPs treatment (Fig. 5) (Table S5). A pairwise comparison of T:R of metals with control showed a significant difference between bacterial communities in the respective planktonic phases (Wilcoxon rank-sum test, $p = 0.007$).

In the follow-up experiment, we found a significant difference in HGT frequency among all treatments. The HGT in PS + Metals was significantly higher than PS and chitosan (Post hoc Wilcoxon test, $p < 0.05$) (Fig. S4). While comparing the two phases, HGT frequency was significantly higher in PS + Metals (particle phase) than in the planktonic phase of the same treatment as well as metal treatment.

3.5. HGT in the natural microbial community of a lake, and temperature effects

The donor strain carrying plasmid pKJK5 was incubated with bacteria from Lake Stechlin at two different temperatures, 20 and 25 °C, in the presence of TWPs. These two temperatures were chosen as they represent the typical temperature range of northern temperate lakes, including Lake Stechlin (Maberly et al., 2020). The fluorescent microscopy images confirmed the colonization of the TWPs and HGT on the surface of the particles by lake bacteria (Fig. 6). The plasmid transfer frequency (T:R) measured on TWPs at 20 °C ($1.37 \pm 0.062 \times 10^{-3}$, mean $\pm$ SD) were 49.7 times higher than those of planktonic bacteria in the same treatment ($2.74 \times 10^{-5} \pm 3.7 \times 10^{-5}$, mean $\pm$ SD) and significantly higher than the control treatment, without TWPs ($p = 0.005$) (Fig. 7). At 25 °C, HGT frequency (T:R) ($120 \pm 60 \times 10^{-3}$, mean $\pm$ SD)

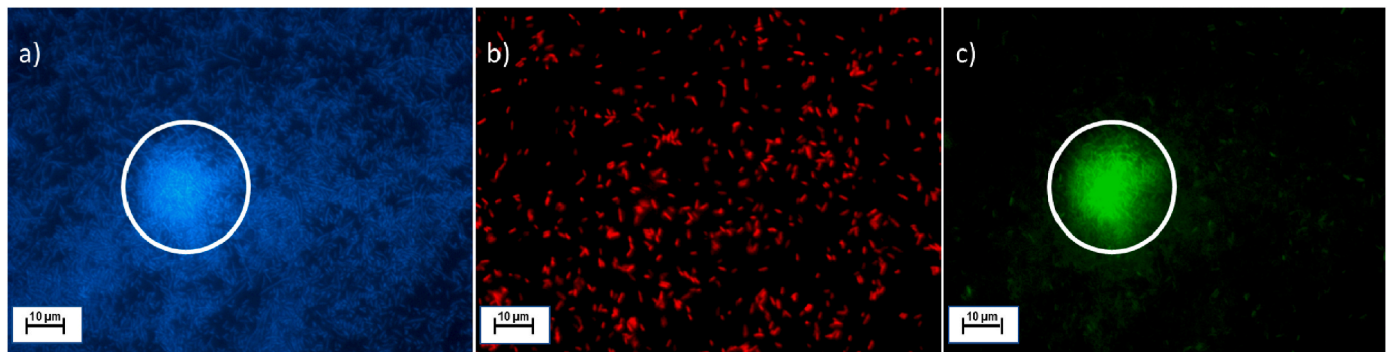


Fig. 4. (a) represents total cells stained by DAPI, (b) represents only the donor cells (*E. coli*) expressing red fluorescence *mCherry-Km^R* gene and (c) represents the transconjugants expressing *gfp* gene fluorescence at 30 °C in the presence of TWPs. The encircled zone in the lower panel shows TWPs as hot spots of biofilm formation and gene exchange.

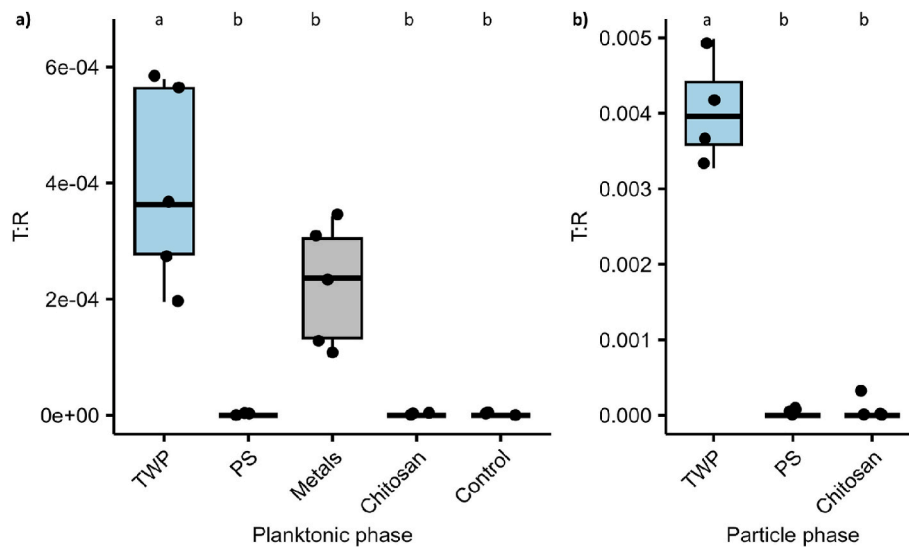


Fig. 5. Box plots and dots represent T:R ratios, which is the representation of HGT in (a) planktonic phase of TWPs, PS, metals, chitosan and control and (b) particle phase of TWPs, PS and chitosan after 24 h of incubation at 30 °C. Error bars represent the standard deviation from the measurements. Different colors depict distinct treatments. Letter 'a' indicates significantly different groups in the HGT frequency between treatments (Post hoc Wilcox test, $p < 0.05$), and 'b' represent non-significant groups.

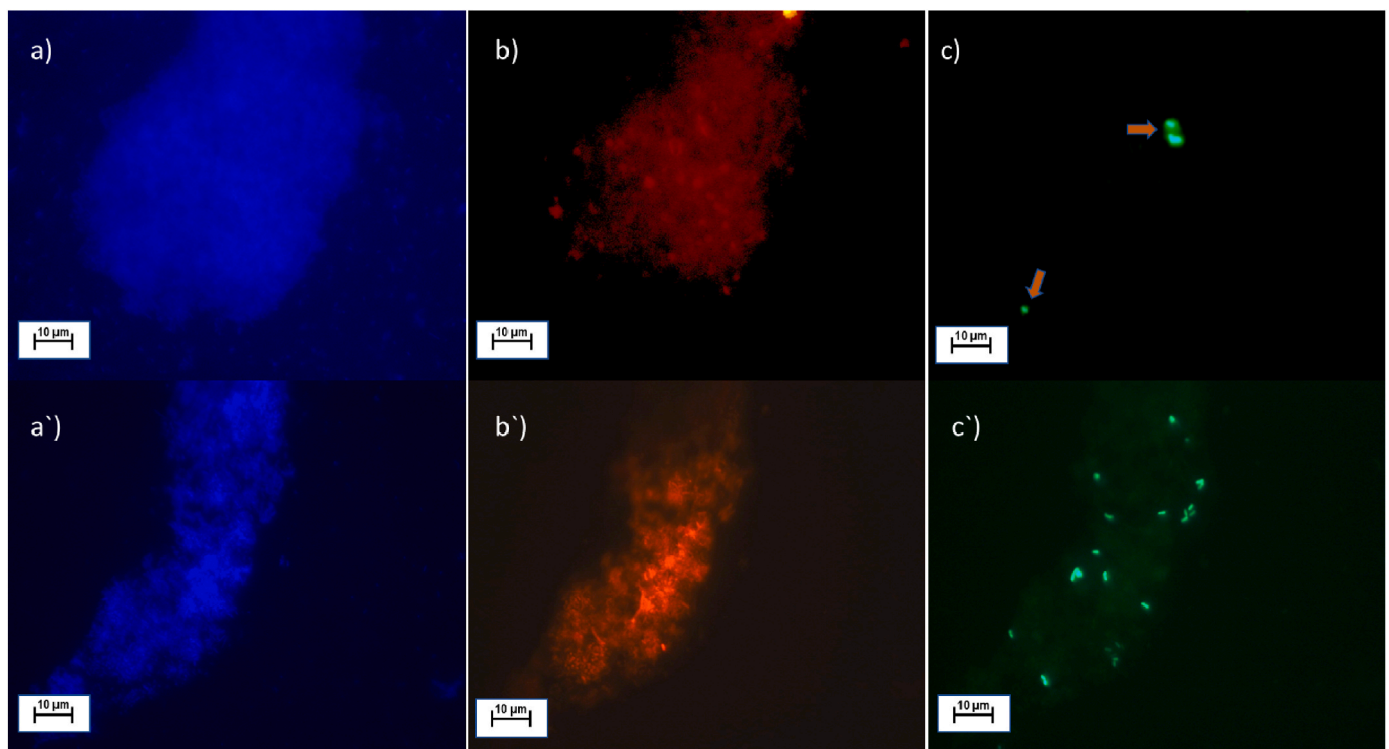


Fig. 6. Biofilm formation and antibiotic resistance gene exchange on TWPs with a natural lake microbial community as recipient. Top panel (a, b, c) represents cells grown at 20 °C. a) represents total DAPI cells stained, b) represents donor cells (*E. coli*) expressing red fluorescence *mCherry-Km^R* gene and c) the transconjugants expressing *gfp* gene fluorescence. Similarly, the lower panel (a', b', c') corresponds to cells grown at 25 °C.

in particle phase was significantly higher than those of planktonic bacteria in the same treatment and the control treatment without any TWPs ($p = 0.005$). HGT transfer frequency (T:R) on particle and planktonic phase at 25 °C was 87.6 and 0.32 times higher than at 20 °C. There was a significant difference in HGT frequency between the particle phase at 20 °C and 25 °C ($p = 0.006$). However, no significant difference was found in the planktonic phases at 20 °C and 25 °C ($p = 0.362$) (Fig. 7).

4. Discussion

Here, we conducted a study to investigate the potential effects of TWPs, which are subject to uncontrolled production and release into the environment, on the transfer of AMR encoding plasmids and hence both, ecosystem and, potentially, human health. Understanding the mechanism of HGT is beyond the scope of this study. Instead, our focus is on investigating the factors that influence HGT on the surface of TWPs. This

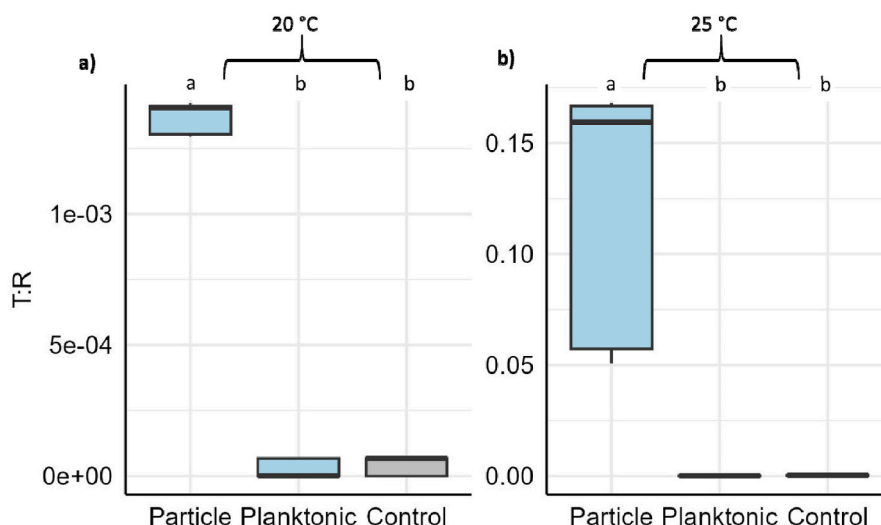


Fig. 7. Box plots represent HGT frequency (T:R) in presence of TWPs at 20 °C (a) and 25 °C (b) with the natural Lake Stechlin microbial community as recipient. HGT is depicted on particle and planktonic phases of TWPs and compared to the control without any particle. Error bars represent the standard deviation from the measurements.

study not only elucidates the impact of MPs on HGT but also provides procedures for conducting similar studies in the future. During the optimization of the transfer assay, the ratio of D/R cells was found to have a significant impact on gene transfer efficiency, indicating the importance of an appropriate balance for successful HGT within a certain time range. Notably, the 1:10 and 1:100 ratios exhibited significantly higher transfer frequencies. Interestingly, the higher number of recipient cells (*Pseudomonas* sp.) on TWPs at ratios of 1:10 and 1:100 suggests that recipient bacteria have a greater propensity to colonize the MPs surfaces compared to the donor cells. Ratios of donors to recipients less than 1:10 reflect environmental conditions where many recipients and fewer potential donors are entering the system (Pallares-Vega et al., 2021) such as activated sludge and to psol (Manti et al., 2008). Previous studies investigating the HGT on MPs may have overlooked the potential influence of the D/R ratio on transfer frequency (Hu et al., 2022; Yuan et al., 2022) and thus comparison of results should be performed carefully. In contrast, Zha et al. (2022) found that higher D/R ratios (4:1) resulted in more gene transfer events due to a better chance of donors to meet recipients. Understanding these dynamics provides valuable insights into the factors influencing gene transfer events on MPs and contributes to our overall understanding of HGT in environmental settings.

Linear regression model analysis indicated that none of the interaction terms (Incubation Time * Temperature, Incubation Time * Ratio, Temperature * Ratio, and Incubation Time * Temperature * Ratio) demonstrated a significant effect on HGT frequency (p-value > 0.05). This suggests that, within the tested conditions, the combined influence of these factors may not have a strong impact on plasmid transfer rates. However, individual variables such as incubation time and temperature were found to have a significant impact on HGT frequency. The lack of significant interaction effects in our model could reflect the complexity of biological systems or the limitations of the experimental setup. Further research may be required to explore more complex or nonlinear relationships between these variables or to apply alternative statistical methods to reveal potential interactions that were not captured in this study.

Previously, HGT in the presence of MPs has been observed at both low temperatures (≤ 25 °C) (Arias-Andres et al., 2018b; Yuan et al., 2022) and high temperatures (> 25 °C) (Zha et al., 2022). The conjugation experiment we conducted in the presence of TWPs at 17 °C, 20 °C, 25 °C, 27 °C, and 30 °C covers the temperature range in the majority of

global freshwater environments (Maberly et al., 2020; Sharma et al., 2015). The highest HGT frequency occurred at 30 °C and decreased with lowering temperatures. The temperature-dependent variation in plasmid exchange is a complex interplay between temperature and microbial dynamics, which can significantly impact the frequency and efficiency of HGT. This could be attributed to microbial metabolism, including the processes involved in HGT, which is often temperature sensitive. The effect of temperature on bacterial conjugation varies depending on the type of plasmid involved (Frost, 2010). For many plasmids, conjugation efficiency decreases at higher temperatures. For example, the IncHI1 plasmid R27 shows optimal transfer between 22 and 30 °C, but transfer becomes undetectable at 37 °C. Temperature influences the expression of transfer genes; in R27, six transfer promoters are repressed at 37 °C compared to 30 °C (Forns et al., 2005; Frost, 2010). Some plasmids, like R27, have temperature-sensitive regulators such as H-NS and Hha orthologs, which repress transfer genes at elevated temperatures (Forns et al., 2005). IncP-1 plasmids use repressors like KorA, KorB, and TrbA to regulate transfer gene expression independently of host functions (Bingle and Thomas, 2001). For these plasmids, the optimal conjugation temperature is around 37 °C, with reduced transfer at both higher and lower temperatures, likely due to temperature-sensitive regulation of transfer genes and protein stability (Frost, 2010). Additionally, temperature affects biofilm formation, which is essential for conjugation. For example, *Salmonella* forms strong biofilms under optimal conditions (37 °C, pH 7.0, and 0.25% glucose) (Roy et al., 2021). Higher temperatures can stimulate the metabolic activity of bacteria, increasing their overall growth rate and replication (Price and Sowers, 2004). Moreover, temperature can affect the formation and stability of biofilms on MPs (Else et al., 2003), which play a crucial role in HGT. Elevated temperatures promote better biofilm formation (Else et al., 2003), creating favorable conditions for gene exchange within biofilms. It was also reported that elevated temperatures can increase the mobility of bacterial cells, facilitating encounters between donor and recipient cells and, subsequently, gene transfer events. Furthermore, conjugation was predominantly observed on TWPs rather than in the surrounding medium, except at 30 °C. This suggests that transfer of genetic material through conjugation is more prominent when bacterial cells are in direct contact with particles rather than freely suspended in the medium.

We also observed that gene transfer frequency was higher after 24 h of incubation as compared to 48 h. This could be attributed to a lower

chance of interaction between donor and recipient cells in the later stage, as the concentration of donor cells decreased significantly (approximately 87.26%) on the particles after 48 h of incubation. Regardless of the duration of incubation, it was consistently observed that the samples treated with TWP exhibited a higher HGT frequency compared to control samples without any particles.

It has been found that MPs selectively enrich the bacterial assemblages and exacerbate the horizontal transfer of ARG transfer through conjugation (Arias-Andres et al., 2018b; Wang et al., 2020). Plasmid *pKJK5:gfpmut3* in *E. coli* MG1655 showed a high permissiveness on the surface of TWP with transfer frequency of $4.5 \pm 1.3 \times 10^{-3}$ (mean $\pm$ SD). Upon microscopic examination of the samples (Fig. 4), it became evident that the majority of the gene transfer activity occurred on the particles rather than in the suspension phase (planktonic).

It is important to note that the environment contains numerous particulate materials that may potentially interfere with the effects of tire microplastics on HGT. However, it is impractical to investigate the impact of all particulate materials within the complex environmental samples. Therefore, in this study, we employed a simplified system by utilizing chitosan as a model particulate material. There was a significant difference in HGT frequency among TWP, PS and chitosan with TWP showing the highest gene transfer. Our findings are in line with a previous study (Zha et al., 2022), which reported that PS MPs with a radius of 500 nm yielded no effect on transfer of the RP4 plasmid in *E. coli* DH5 α . It is important to note that Zha et al. (2022) conducted gene transfer experiments using filter-mating, while our study focused on monitoring HGT in a liquid suspension medium vs. particles, which is more relevant to natural aquatic systems. It signifies the conjugative transfer of ARG by MPs is size-dependent (Zha et al., 2022) and in the case of TWP it isn't only the polymer type that enhances gene transfer, additional factors seem to play a role.

It is not surprising that Zn was the highest of all measured metals in TWP. The combination of metals found are consistent with previous studies (Adeyi and Oladoye, 2020; Adachi and Tainosho, 2004). During the vulcanization process, zinc oxide (ZnO) is injected as an activator and thus constitutes 0.4%–4.3% of the finished tire tread. Our findings suggest that TWP not only act as a surface for bacterial attachment but are also a significant source of metals for HGT. This phenomenon was more pronounced when metals were associated with MPs than in suspension (planktonic phase). Whether the leached metals or metals bound to TWP played a role in HGT requires further investigation. Metals, particularly heavy metals such as Zn and Cd, can act as strong environmental, selective pressures in microbial communities, which can promote the survival and proliferation of metal-resistant bacteria or mobile genetic elements also carrying metal resistance genes (Liu et al., 2022; Chen et al., 2023). Metal resistance genes and antibiotic resistance genes are often found together on the same mobile genetic elements (Li et al., 2016). Consequently, exposure to heavy metals can contribute to the selection and dissemination of antibiotic resistance in bacteria through a co-selection mechanism. This co-selection phenomenon occurs because many ARG are physically linked to genetic elements that also confer resistance to heavy metals, such as plasmids or transposons. For example, the plasmid *pAFS11* from *Staphylococcus aureus*, which carries the methicillin resistance gene, demonstrates the co-occurrence of multiple antimicrobial resistance genes (such as tetracycline, kanamycin, apramycin, trimethoprim, and macrolides) alongside heavy metal resistance genes for copper (Cu) and cadmium (Cd) (Slifierz, 2016).

Since TWP comprise a complex blend of different compounds, our study partially explained the role played by metals in HGT. However, we suggest that other components like elemental carbon (black carbon), polycyclic aromatic hydrocarbons that are also essential components in tires (Aatmeeyata & Sharma, 2010) might also play a role for HGT in the presence of TWP. Further research is required to know the potential contributions of other tire-derived components and the specific impact of bound versus leached metals needs deeper exploration.

We observed significant HGT rates in the natural lake microbial community (Lake Stechlin, Germany) in the presence of TWP. The increase in HGT frequency in the lake community in presence of TWP and at higher temperatures closely mirrored what we observed in our controlled laboratory settings. In the laboratory, clean TWP were allowed for metal leaching and bacterial attachment, while in nature, leaching may occur over longer periods and in larger volumes, potentially limiting metal concentrations above gene transfer thresholds. The presence of TWP in natural ecosystems may interact with various abiotic (e.g., UV radiation and redox changes) and biotic processes (e.g., biofouling), which potentially influence HGT differently compared to laboratory conditions. While there are certain reports of very high concentrations of TWP, typical concentrations are lower than what we have used in the lab. Despite the background of natural particles, previous research has demonstrated that anthropogenically introduced surfaces like MP and TWP can significantly enhance gene transfer compared to natural particles (Weise et al., 2022).

It is also important to note that the higher nutrient concentrations in lab conditions, compared to natural lake water samples, can potentially positively influence HGT (Arias-Andres et al., 2018b). Surprisingly, the observed HGT frequency in a complex lake microbial community was multiple times higher than those observed in a single microbial strain as recipient. Monitoring HGT in the environment is difficult because the majority of indigenous bacteria remain uncultivated (Amann et al., 1995). However, an effective approach to study HGT in complex environments without relying on cultivation is the use of fluorescently labelled strains carrying specific plasmids (Sørensen et al., 2005a). This method shows promise in overcoming several methodological challenges posed by cultivation biases in environmental HGT studies (Arias-Andres et al., 2018b).

In our lab experiments, we maintained a constant and controlled environment with stable temperature, pH, and nutrient availability, which clearly differ from naturally fluctuating conditions found in lake environments. Therefore, the transferability of lab data to field experiments has severe limitations and thus complicates any direct comparison. Nevertheless, conducting an experiment under controlled conditions allowed us to assess whether a specific factor (heavy metals) can influence HGT in the presence of TWP. It also allowed us to investigate whether TWP facilitate HGT in a complex natural bacterial community. Therefore, while our study provides valuable insights into the potential role of TWP in mediating HGT, caution is warranted in directly extrapolating our lab findings to project HGT in natural environments without any further research and consideration of these complexities.

In our previous studies, we have observed a transfer frequency of 2.6×10^{-4} (T:R) in Lake Stechlin by incubating PS strips at 20 °C (72 h incubation; Arias-Andres et al., 2018b). A study in river water reported a T:R ratio of $\sim 1.6 \times 10^{-4}$ at 20 °C after 96 h of incubation (Muela et al., 1994). In this study, the HGT frequency on TWP with a single recipient *Pseudomonas* sp. was found to be in the order of 10^{-3} at 30 °C while in the natural microbial lake community, the transfer frequency reached a magnitude of up to 10^{-1} even at low temperature (20 °C). This demonstrates the increased permissiveness of the ARG carrying plasmid in the presence of TWP, particularly in the natural environment. Another factor might be the presence of microbial populations in the community that are more closely related to the donor species than *Pseudomonas* sp. strains isolated from the same environment (Alderliesten et al., 2020). Eventually, due to the higher HGT frequency due to TWP has many environmental complications. HGT significantly impacts the environment by contributing to the spread of antibiotic resistance genes (ARGs). This poses risks to ecosystems and public health by leading to drug-resistant infections that are difficult to treat and often require stronger, more expensive, or toxic medications. HGT can also increase bacterial pathogenicity by transferring virulence factors. Additionally, biofilms, which facilitate HGT, thrive in various environments, further enhancing gene exchange.

TWPs like other MP types provided congenial surfaces for biofilm formation. Within biofilms, bacteria are situated in close proximity, increasing the likelihood of intense cell-to-cell interactions. Our previous study (Arias-Andres et al., 2018b) has confirmed that such interactions on the surface of microplastics (MPs) lead to HGT. As a result, niches with elevated plasmid transfer rates between neighboring cells can form (Weise et al., 2022). SEM (Fig. 2) and fluorescence MPs (Figs. 4 and 6) validate that TWPs were indeed covered in biofilms that contained donor bacteria and increased numbers of green fluorescent transconjugant cells with the plasmid. Consequently, TWPs provide bacteria with novel surfaces for biofilms, potentially promoting bacterial plasmid transfer and, subsequently, the spread of antimicrobial resistance.

5. Conclusion

We have performed a comprehensive study of colonization and HGT on TWPs. Our findings suggest that TWPs pollution, especially in environments that may harbor pathogenic bacteria, such as wastewater, could serve as an efficient substrate for HGT of, e.g., antibiotic resistance among different bacteria. This raises significant concerns regarding the potential for the spread of antibiotic resistance in such environments, which could have serious implications for public health and the effectiveness of antibiotic treatments. The variations in gene transfer (HGT) and biofilm formation among tested materials could impact microbial communities in TWPs-affected aquatic environments. While the presence of metals in TWPs facilitates the exchange of ARGs in laboratory experiments, the influence of TWPs on HGT may vary in natural aquatic environments due to interactions with abiotic and biotic factors, such as UV radiation, redox potential, and nutrient availability. Future research, including quantum-chemistry-based dynamic models, is needed to elucidate these interactions and their potential consequences for human and environmental health.

The observed temperature effects on HGT in presence of TWPs emphasizes the significance of comprehending the implications of global warming. Future studies should encompass a wide range of aquatic ecosystems characterized by diverse temperatures to gain a comprehensive understanding of the complex relationship between temperature, MP types, and HGT processes. Such investigations will pave the way to assess and address the potential environmental consequences of climate change and MPs pollution. Investigating contaminants such as MPs in natural conditions poses significant challenges. Therefore, many studies resort to laboratory-based investigations to mimic natural conditions. While these laboratory setups may not fully replicate the complexity of natural scenarios, they do provide valuable insights into the potential factors influencing processes, as observed in our study on HGT in the presence of TWPs in water samples collected from the lake.

CRedit authorship contribution statement

Yousuf Dar Jaffer: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fazel Abdolapur Monikh:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Klümper Uli:** Writing – review & editing, Supervision, Methodology, Investigation. **Hans-Peter Grossart:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2024.120187>.

Data availability

No data was used for the research described in the article.

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